

A Spiking Neural Network Approach for EEG-Based ADHD Classification Using Theta Band Power

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Abstract—Early detection of Attention-Deficit/Hyperactivity Disorder (ADHD) is crucial for effective intervention. However, traditional diagnostic methods are mostly subjective and take a lot of time. This shows the need for more objective and efficient ways of diagnosis. This study investigates the use of spiking neural networks (SNNs) to classify Attention-Deficit/Hyperactivity Disorder (ADHD) and control subjects with Electroencephalography (EEG). The EEG dataset was obtained from IEEE DataPort and was preprocessed to extract theta-band power (4–8 Hz). The signals were then encoded into spikes using Ben’s Spiker Algorithm (BSA) and mapped to Talairach brain coordinates, creating a three-dimensional spiking neural cube (SNNcube) within the NeuCube framework. The classification was done with the dynamic evolving spiking neural network (deSNN). The model reached an overall accuracy of 67.57% on hold-out dataset and a Leave-One-Out Cross Validation (LOOCV) accuracy of 57.02%, capturing spatio-temporal EEG patterns related to ADHD. These results suggest that SNN-based frameworks like NeuCube may be valuable for neuroinformatics applications such as brain-computer interfaces (BCIs) and attention-monitoring systems, where understanding temporal brain activity is important.

Keywords—Electroencephalography (EEG), Attention-Deficit/Hyperactivity Disorder (ADHD), Spiking Neural Networks (SNN), NeuCube, Theta-Band Power, Machine Learning, Classification.

I. INTRODUCTION AND RATIONAL

Attention-Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental condition that affects about 2 to 5% of the global population [1]. It often continues into adulthood and leads to significant cognitive and behavioural challenges. Early diagnosis is key for effective intervention. However, traditional clinical assessments are mostly subjective and take a lot of time.

Electroencephalography (EEG) has become a useful tool for measuring brain activity directly and non-invasively, offering excellent temporal resolution. Previous studies have shown that ADHD is linked to alterations in EEG spectral patterns, particularly with higher theta-band activity in frontal and central brain regions [2]. This observation provides a motivation to explore theta power as a potential feature for classifying individuals with ADHD and control subjects. Recent progress in machine learning and Artificial neural network has further enhanced the analysis of EEG data, allowing researchers to move beyond traditional approaches. Applying these algorithms to theta-band power enables a detailed classification of ADHD and control groups, contributing to improved understanding of ADHD.

The chosen topic is highly relevant as it brings together neuroinformatics, which involves EEG data processing and Artificial Intelligence through Spiking Neural Network modelling. This combination offers insight into ADHD-related brain activity and practical assessment of tools for classification.

The remainder of this report is organized as follows. Section II presents the *review of existing literature*. Section III describes the *Spatio-temporal Dataset*, outlining data feature characteristics. Section IV details the *Data Analysis* which includes preprocessing, spike encoding, and SNNcube training. Section V presents the *Results and Discussion*, followed by Section VI, which outlines the *Practical Applications* of this research. Section VII compares the performance of the NeuCube model with traditional machine learning results obtained using the NeuCom software. Finally, Section VIII concludes the report and summarizes the main findings, followed by the *References* section.

II. LITERATURE REVIEW

Electroencephalography (EEG) has become a non-invasive and cost-effective method for studying brain waves related to ADHD. It offers high-resolution insights into brain activity. Researchers have consistently found abnormalities in EEG spectral bands, especially increased theta power in people with ADHD [2]. These findings suggest that these electrophysiological markers could be useful for accurate classification. Clarke et al. [2] conducted a key comparative EEG study on ADHD subtypes. They looked at the differences in spectral power between children diagnosed with ADHD and those without ADHD. Their findings showed that children with ADHD consistently had higher theta activity (4–8 Hz) compared to non-ADHD individuals. This increased theta activity was suggested as a possible marker for ADHD [2]. These results gave support for computational models to use theta-band features as indicators for automated ADHD classification.

Machine learning techniques are being used more to analyse EEG. They help to classify the differences between ADHD and control groups. Traditional classifiers like support vector machine (SVM), K-nearest neighbours (kNN), and multilayer perceptron (MLP) have been trained on EEG features, including theta power and theta/beta ratios. For instance, XGBoost was used on 19 channel EEG band features and reached 90.81% accuracy with leave-one-subject-out cross validation. SHapley additive exPlanation (SHAP) was also performed to find important contributing features [3].

Early studies on EEG-based ADHD classification mostly used traditional ML methods. Researchers extracted spectral features, such as the Statistical features, Entropy measures, Hjorth parameters to distinguish ADHD from control groups [4]. While these models were clear, their reliance on feature engineering and their sensitivity to noise had some limitations.

Deep learning (DL) methods, such as Convolutional Neural Networks (CNNs) have shown good results in classifying ADHD from EEG data. However, these models require significant computational resources and are biologically abstract since they depend on continuous-valued activations that do not accurately reflect how neurons communicate through spikes. This has led to interest in Spiking Neural Networks (SNNs), a newer type of neural architecture that processes information as discrete spike trains, closely resembling the way biological neurons communicate [5]. SNNs have several advantages over traditional neural networks, including event-driven computation, high temporal precision, and energy efficiency, especially when used on neuromorphic hardware [5], [6]. SNNs can offer a biologically realistic and efficient way to automate ADHD detection and study related neurocognitive disorders.

The foundational work by Kasabov, “Spiking Neural Networks: Background, Recent Development and the NeuCube Architecture,” gives a basic understanding of how spiking neural networks (SNNs) work and their relevance to processing spatiotemporal data. The NeuCube architecture simulates EEG signals in a 3D spiking reservoir. This setup allows for unsupervised learning of temporal patterns and supervised mapping of outputs. Its bio-inspired design makes it easier to interpret results and improves efficiency in modelling temporal sequences [5].

Cai et al. in “Spiking Neural Networks for EEG Signal Analysis: From Theory to Practice” presented a detailed framework for SNN-based EEG analysis. They explained encoding methods such as rate and temporal coding, highlighted neuron models like Leaky Integrate-and-Fire (LIF) and Izhikevich and discussed how SNNs can be implemented on neuromorphic processors, including Intel Loihi and IBM TrueNorth. Their research emphasizes that SNNs are well-suited for EEG analysis due to their event-driven computations and their ability to handle complex neural data in real-time [7].

Similarly, Kim and Kim investigated the biomedical applications of SNNs, pointing out their benefits in energy efficiency, real-time results, and temporal resolution. They mentioned that while SNNs have shown to be effective in processing biomedical signals like EEG, EMG, and ECG, their application in clinical diagnostics, such as for ADHD, is still not fully explored, but it holds great potential [6].

Among empirical studies, Pedrollo et al. [8] made the first attempts to apply SNNs directly to ADHD classification. They used resting-state EEG data from the Healthy Brain Network dataset. Their LIF-based SNN, which used threshold-based temporal encoding, distinguished ADHD subtypes. The model achieved 72% accuracy in classifying ADHD versus control and 68% for subtype differentiation. It outperformed traditional ML algorithms like SVM and Random Forest by about 9%. This study showed that SNNs can effectively capture temporal spiking patterns relevant to ADHD diagnosis.

Other EEG-related SNN studies provide more insight into the flexibility of spiking architectures. For example, Jain et al. proposed a Fuzzy Logic Spiking Neural Network (FLSNN) for diagnosing multiple disorders such as epilepsy, Parkinson’s, Alzheimer’s, schizophrenia, and stroke. They integrated fuzzy reasoning to handle uncertainty in EEG signals and improve interpretability. Their framework achieves 97.46% accuracy in binary classification and 98.87% accuracy in multi-class classification [9]. Similarly, Li et al. developed a spatial SNN model for testing concealed information using EEG, showing high accuracy of 90.15% in classifying event-related potentials (ERPs) by taking advantage of spatial spiking patterns [10]. These studies highlight how SNNs can adapt to various EEG classification tasks beyond ADHD, reinforcing their potential as a unified computational framework for neurocognitive analysis.

A review of current literature shows that using Spiking Neural Networks (SNNs) for EEG-based ADHD detection is mostly unexplored. Among the studies examined, only one research project conducted by Pedrollo et al. [8] has directly applied an SNN model for classifying ADHD. This lack of research points to a significant gap in the field. It suggests that using biologically inspired spiking architectures in ADHD analysis is still very new. As Cai et al. [7] and Pedrollo et al. [8] point out, obstacles like small datasets, complex training processes, and a lack of standard methods for spike encoding limit their broader application.

The reviewed literature shows a shift from feature-based machine learning to deep learning and, later to spiking neural networks. Traditional machine learning methods provide clear explanations but rely heavily on manually created features. Deep networks achieve better accuracy but lack biological realism. Spiking neural networks combine a realistic approach to neurophysiology with efficient computation [5], [6], [7]. The limited number of SNN-focused ADHD studies highlights a clear research gap and offers a valuable opportunity to contribute new insights into the field.

III. SPATIO-TEMPORAL DATA

A. Data Collection

The dataset used in this study is the EEG Data for ADHD/Control Children, published on IEEE DataPort by Ali Motie Nasrabadi et al [11]. The data was collected in collaboration with clinical psychologists and neurologists at Shahid Beheshti University of Medical Sciences in Iran. EEG recordings were taken from 121 children aged 7 to 12 years. This group included 61 individuals diagnosed with Attention-Deficit/Hyperactivity Disorder (ADHD) and 60 control individuals.

Each child took part in a visual attention task focused at triggering attention-related brain responses. During the task, children looked at images of cartoon characters, with 5 to 16 characters per image, and were asked to count the number of target characters shown. EEG signals were recorded continuously throughout this cognitive task. The 19-channel EEG cap following the international 10-20 electrode placement system as shown in figure 1 was used, with a sampling rate of 128 Hz [11].

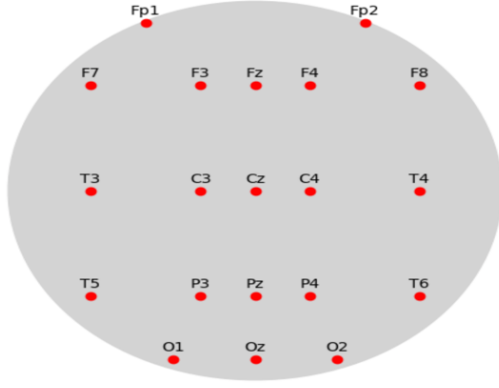


Figure 1: Electrode Location According to International 10-20 System for EEG.

The dataset includes multichannel EEG signals. Each channel corresponds to a specific location on the scalp. This setup captures brain activity from different areas of the cortex. The spatial part of the data is represented by 19 electrodes: Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2. These electrodes detect electrical potential differences linked to the frontal, central, parietal, and occipital regions of the brain that are involved in attention and control [11].

B. Class Distribution

For this study, the dataset was divided into two distinct classes:

- i. ADHD group, EEG recordings from 61 children who were clinically diagnosed with Attention-Deficit/Hyperactivity Disorder. This diagnosis was confirmed through behavioural assessments and expert evaluation.
- ii. Control group (HC), EEG recordings from 60 typically developing children with no clinical diagnosis of ADHD or other neurodevelopmental disorders.

The class distribution in Figure 2 displays the number of ADHD and control participants. The dataset is balanced between the two groups, which helps prevent the model from favouring one class. These class labels are clearly provided in the target file for classification tasks. This results in a balanced class distribution suitable for classification tasks.

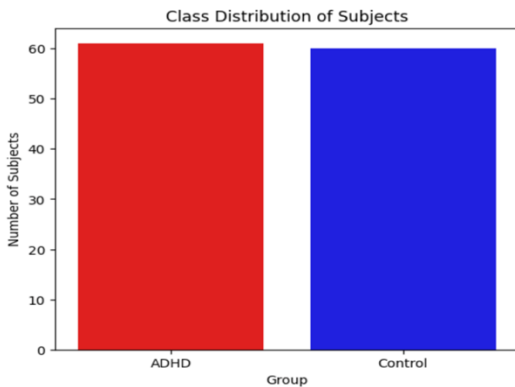


Figure 2: Class Distribution

C. Previous Research on the dataset

The EEG data for ADHD/Control children, is widely used in research for automated ADHD detection. Recent studies have used this dataset to assess both traditional machine learning and deep learning models.

In one of the studies with this dataset, Alim et al. developed a computer-aided diagnosis system for automatically identifying ADHD in children. The authors extracted statistical and time-frequency features from EEG signals and trained Gaussian support vector machines classifier. The support vector machine classifier achieved a classification accuracy around 94% [4]. However, relying on handcrafted features limited the ability to capture deeper spatiotemporal details in EEG data, leading to further research into deep learning options.

Loh et al. [12] proposed ADHD/CD-NET, an explainable deep neural network framework. It aims to characterize ADHD and its comorbidity with Conduct Disorder (CD). Their study used both the public Nasrabadi dataset and a private clinical dataset that included 123 multi-class participants: 45 with ADHD, 62 with ADHD+CD, and 16 with CD. EEG signals first underwent preprocessing through Continuous Wavelet Transform (CWT) and were then converted into channel-wise correlation matrices. These served as input for a two-dimensional Convolutional Neural Network (CNN). This model reached 93.7% accuracy for multi-class classification and up to 98.19% accuracy on the binary ADHD versus control task using the public dataset. To improve clinical transparency, the authors added Gradient-weighted Class Activation Mapping (Grad-CAM). This allowed for the visualization of EEG channels that significantly influenced the model's decision-making [12].

Later Barua et al. [13] introduced a hybrid deep learning model that combined autoencoder-based feature extraction with a Residual Network (ResNet) improved by a Double Augmented Attention Mechanism (DAAM). Using the same EEG dataset, their model employed the autoencoder to clean and compress EEG representations before using the ResNet-DAAM for classification. This architecture achieved an impressive accuracy of over 99% [13]. The study showed the potential of hybrid and attention-driven deep models to improve both accuracy and understanding, highlighting the dataset's value for testing ADHD detection algorithms.

Together, these studies demonstrate a clear change in how the Nasrabadi EEG dataset is used. They have shifted from traditional machine learning models [4] to attention-based deep learning frameworks [12], [13]. The continued use of this dataset in various studies highlights its importance as a standard for EEG-based ADHD classification research.

IV. DATA ANALYSIS

A. Data Preprocessing

To ensure data quality and extract meaningful features for ADHD classification, all EEG recordings were processed using a structured pipeline developed in Python. The preprocessing was done to remove noise, isolate relevant frequency, and compute key theta-band features linked to ADHD-related activity. The complete preprocessing workflow is given in the following steps:

- i. **Bandpass Filtering:** EEG signals naturally contain unwanted low-frequency drifts, such as those caused by perspiration or electrode movement, and high-frequency noise from muscle artifacts or electrical interference. To reduce these effects, a 4th-order Butterworth bandpass filter with cutoff frequencies between 0.5 Hz and 45 Hz was applied to each channel. Butterworth design was chosen for its flat frequency response, which ensures smooth reduction at the boundaries without causing sharp distortions. This frequency range effectively preserves the main EEG bands, including delta, theta, alpha, and beta. The filtered output represents clean, physiologically valid brainwave activity suitable for further analysis [14].
- ii. **Segmentation into Temporal Windows:** After filtering, each continuous EEG recording was divided into non-overlapping 1-second windows, which correspond to 128 samples per segment at the 128 Hz sampling rate. This division aimed to simplify the dataset's computational demands and enable better analysis. Each EEG session included about 3 to 4 minutes of continuous recording per subject. Loading the entire signals into the NeuCube environment was computationally heavy and took a long time. Breaking the signals into shorter segments not only cut down processing time but also allowed for more efficient data management within the NeuCube framework, all while keeping the key temporal features intact.
- iii. **Theta-Band Power Computation:** For each 1-second window and electrode channel, the Power Spectral Density (PSD) using Welch's method was computed. This method gives a smooth estimate of how power is distributed across frequencies. From the PSD, the theta-band power (4–8 Hz) was extracted by adding up the spectral values within that frequency range.
- iv. **Feature Matrix Construction and Storage:** The computed theta power values were stored into a matrix for each subject. This preserved both the spatial and temporal features of the EEG data. Each matrix was saved as a separate CSV file in an output directory. This step was necessary for loading the data into Neucube software [15].

B. Spike Encoding

After finishing the preprocessing phase, the extracted theta-band EEG were imported into the NeuCube framework for analysis and classification. NeuCube is a computational system that mimics brain-like information processing. It represents neural activity in three dimensions and encodes temporal signals as discrete spikes. This platform allows visualization, training, and interpretation of EEG in a way that makes biological sense [15].

The continuous EEG signals were converted into spike trains using Ben's Spiker Algorithm (BSA) and were compared with the Threshold-Based Representation (TBR) approach. BSA is part of the stimulus estimation family of

methods and works by creating a unipolar spike train from which the original analog signal can be reconstructed by convolving this spike train with a finite impulse response (FIR) filter [16]. This method produces sparse but information-rich spike trains that maintain the temporal structure of the original signal.

In contrast, TBR is a temporal contrast method that creates spikes whenever the change in amplitude between samples goes beyond a specific threshold. While it is computationally efficient, TBR can be sensitive to high-frequency noise and may miss subtle temporal variations. This makes it less suitable for continuous, low-frequency EEG signals. BSA, however, was designed to encode smooth, slowly changing signals. In this study, temporal precision and reconstruction accuracy are vital [16].

C. Initialisation and training of SNNcube

Following spike encoding, the next stage involved starting and training the Spiking Neural Network cube (SNNcube), which forms the main structure of the NeuCube framework. The SNNcube consists of a group of spiking neurons arranged in space, with the input neurons often having specific coordinates. The three-dimensional arrangement of the neurons and their connections need a visualization method that goes beyond what a standard 2D connectivity can offer [5]. Each neuron in the cube matches a location mapped using Talairach brain coordinates. For initialization, all Talairach coordinates were uploaded into the NeuCube environment, along with the electrode coordinates for the 19 EEG channels used in this study. Each input channel was projected into the cube based on its position in the cortex. The SNNcube developed into a 3D brain-like model showing neurons as shown in figure 3.

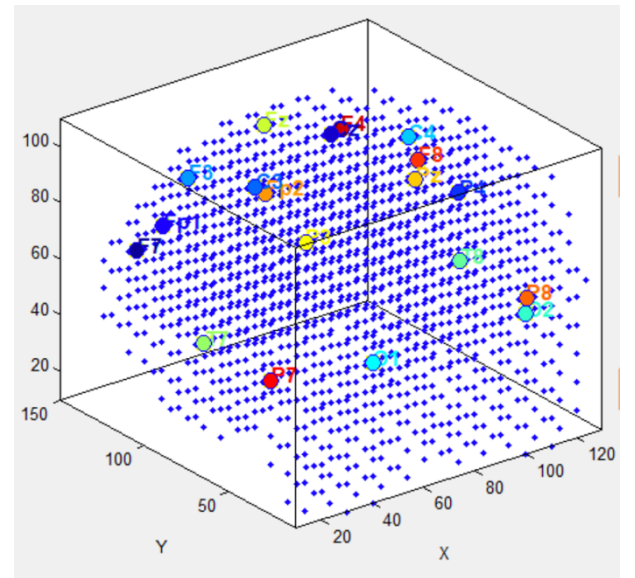


Figure 3: SNNcube showing the 3D-spatial mapping of EEG electrode locations onto Talairach brain coordinates.

After initialisation, the cube was trained to create synaptic connections. The learning process followed the principle of Spike-Timing-Dependent Plasticity (STDP), where the strength of a synaptic connection changes based on the pre-synaptic neuron firing [17].

The SNNcube training parameters were adjusted from the default values to improve the network’s ability to learn from EEG data while ensuring stability and realism. The configuration included a potential leak rate of 0.003, an STDP learning rate of 0.01, a firing threshold of 0.5, a training iteration of 1, a refractory time of 5, and an LDC probability of 0.

The potential leak rate was slightly increased from 0.002 (default) to 0.003 to allow neurons to reset their membrane potentials a little faster. This change helps prevent neurons from holding onto charge for too long. The STDP learning rate stayed the same at 0.01, keeping a balance between learning speed and stability. The firing threshold was maintained at 0.5 because this value offered a good firing frequency. It was low enough for neurons to respond to meaningful patterns and high enough to avoid constant firing caused by noise. The training iteration count remained at 1 to limit computation time. Since EEG data already shows a minute of activity, one iteration was enough to capture key firing relationships. The refractory time was reduced from 6 (default) to 5 to let neurons fire slightly more often after each spike. Finally, the LDC (Lateral Distance Coefficient) probability was kept at 0, meaning that neuron connections were not biased by distance. This ensured that all neurons had an equal opportunity to connect.

After training, the SNNcube developed into a 3D brain-like model showing connectivity patterns between spatially related neurons as shown in figure 4.

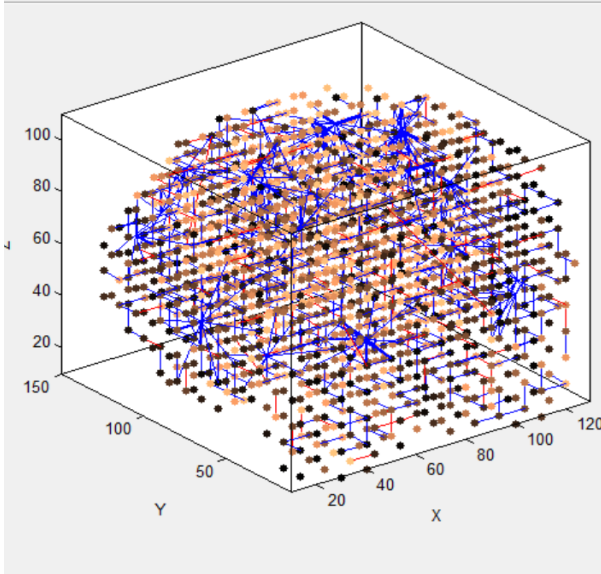


Figure 4: Visualization of the trained SNNcube showing neuron connectivity

D. deSNN Classification

After the SNNcube was complete, classification of the EEG data was carried out using the Dynamic Evolving Spiking Neural Network (deSNN) algorithm. The deSNN is a supervised, event-driven classifier integrated in the NeuCube framework that extends the evolving spiking neural network (eSNN) model proposed by Kasabov *et al.* [5]. It combines Rank-Order (RO) learning with Spike-Driven Synaptic Plasticity (SDSP) to capture both spatial and temporal characteristics of spiking activity. This hybrid learning approach enables the deSNN to utilize the order of the first input spikes for rapid one-pass learning and dynamically

adjust synaptic weights based on subsequent spike timings, resulting in an adaptive and temporally precise classification process [18].

The Dynamic Evolving Spiking Neural Network (deSNN) classifier was tuned to improve its sensitivity to firing patterns learned by the SNNcube. Four main parameters, Mod, Drift, k, and Sigma, were changed from their default NeuCube settings to find a balance between flexibility and generalization during classification.

The Mod parameter was raised from 0.8 to 1.1 to increase the initial impact of early spikes. In the deSNN learning process, early spikes are often the most informative because they indicate strong activations. The Drift coefficient stayed the same at 0.005, allowing for gradual weight decay over time. This decay helps avoid older connections from taking over, keeping the model flexible to adapt to new patterns. The k-value, which represents the number of nearest neighbours, was increased from 3 to 5 to create smoother decision boundaries. By looking at a slightly larger neighborhood, the classifier became less sensitive to outliers and noise, which is important when handling EEG data. Lastly, the Sigma value was increased from 1 to 3, widening the Gaussian activation function used in neuron similarity calculations. This change allows neurons to respond more flexibly to input variations, improving generalization across overlapping EEG patterns.

Overall, these tuning adjustments aimed to improve classification stability and decrease sensitivity to noise. The deSNN model was trained and tested using the 0.7:0.3 training-testing ratio.

V. RESULTS AND DISCUSSION

The NeuCube model was evaluated using two testing approaches: 30% hold-out dataset testing and Leave-One-Out Cross Validation (LOOCV). Both methods were used to assess how accurately the trained model could classify EEG data into ADHD and control groups.

In the hold-out dataset test, 70% of the data was used for training and 30% for testing. The model achieved an overall classification accuracy of 67.57%, with 68.42% accuracy for the ADHD class and 66.67% accuracy for the control class. This result shows that the model could learn and recognize general spatio-temporal firing patterns from both groups fairly well. However, there is still room to improve in distinguishing.

The Leave-One-Out Cross Validation (LOOCV) method was also performed to test how well the model generalizes when each sample is tested individually. Under this evaluation, the overall accuracy was 57.02%, with 59.02% accuracy for ADHD and 55% for control. The lower accuracy compared to the hold-out dataset test indicates that while the model can learn patterns effectively, it may be sensitive to differences across participants — a common challenge in EEG-based studies due to variability in brain activity across subjects.

Overall, the NeuCube framework successfully demonstrated its ability to analyse spatio-temporal EEG data using spike-based learning. Despite moderate accuracy values, the results confirm that the model could distinguish ADHD-related patterns from those of control participants. Further optimization, such as refining spike encoding parameters or adjusting NeuCube’s configuration, may improve performance in future experiments.

VIII. CONCLUSION

This research focused the use of the NeuCube spiking neural network (SNN) framework for classifying Attention-Deficit/Hyperactivity Disorder (ADHD) and control subjects using EEG-Theta band. The EEG dataset, sourced from IEEE DataPort, was preprocessed to extract theta-band power (4–8 Hz). The data were then converted into spikes through Ben's Spiker Algorithm (BSA) and spatially mapped to Talairach brain coordinates to create a three-dimensional spiking neural cube (SNNcube).

The NeuCube model successfully learned spatio-temporal patterns within EEG activity using Spike-Timing-Dependent Plasticity (STDP) and classified samples through the dynamic evolving spiking neural network (deSNN). The model achieved an overall accuracy of 67.57% and Leave-One-Out Cross Validation (LOOCV) of 57.02%. These results demonstrate NeuCube's ability to process brain data.

The findings highlight the potential of spiking neural networks to model real brain activity more closely than conventional machine learning approaches. This approach can be extended to applications like brain-computer interfaces (BCIs), where understanding the temporal evolution of neural activity is essential. While traditional machine learning models like those implemented in NeuCom achieved higher numerical accuracy, NeuCube provides a more clear and biologically understandable framework for studying brain dynamics.

VI. PRACTICAL USES

The EEG analysis done in this study can help develop Brain-Computer Interfaces (BCIs). These interfaces enable direct communication between the brain and external devices. BCIs translate brain signals, usually EEG and ECoG into computer commands. These commands can control external systems like cursors, prosthetic limbs, or assistive devices [19].

In ADHD research, the NeuCube framework and EEG-based spiking neural analysis could create attention-monitoring BCIs that offer real-time neurofeedback. By recognizing neural patterns linked to attention lapses, these systems could alert users or change learning environments on the spot. For instance, they could pause lessons or provide visual reminders when focus decreases. This method can help with cognitive training and therapy for children or adults with ADHD, improving their concentration and self-regulation skills.

Moreover, these BCIs could also be used in education and clinical settings. Continuous monitoring of EEG activity allows for personalized feedback and tracking of treatment progress. By incorporating NeuCube into EEG-based BCIs, we can create adaptive, understandable, and energy-efficient systems that connect neuroscience with real-world assistive technology.

VII. COMPARISON WITH NEUCOM

EEG dataset was applied on both Neucom and Neucube platform to classify ADHD and control participants based on theta-band features. While NeuCom employs conventional machine learning algorithms such as Support Vector Machines (SVM), Multilayer Perceptron (MLP), and Evolving Classification Function (ECF), NeuCube utilizes a biologically inspired spiking neural network (SNN) architecture that processes temporal patterns through spike encoding [15], [20].

In the NeuCom experiments, multiple algorithms were tested. The ECF model achieved the highest hold-out dataset accuracy of 99%, followed by ECMC at 95%, and SVM (RBF kernel, $\gamma = 2$) at 87.6%. The MLP (2 hidden units) and multilinear regression models produced lower accuracies of 76.03% and 74.38%, respectively. However, across all models, Leave-One-Out Cross Validation (LOOCV) results averaging around 58–63%, with SVM (RBF) and ECF performing best at 62.8% [21]. This indicates that the NeuCom models could fit the training data effectively.

In contrast, the NeuCube model, which integrates spatial brain mapping and temporal encoding through Ben's Spiker Algorithm (BSA) and Spike-Timing-Dependent Plasticity (STDP), achieved a hold-out dataset accuracy of 67.57% and LOOCV accuracy of 57.02%. Although slightly lower than traditional classifiers in performance, the NeuCube captures the spatio-temporal evolution of brain signals, rather than relying solely on static theta power features.

Overall, NeuCom's machine learning algorithms achieved higher numerical accuracy, but NeuCube offered greater explainability and biological realism.

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